

AWS

COMPLETE NOTES

= (AWS) =



Cloud Skills Made Simple



✦ @NeerajSingh-DevOps ✦



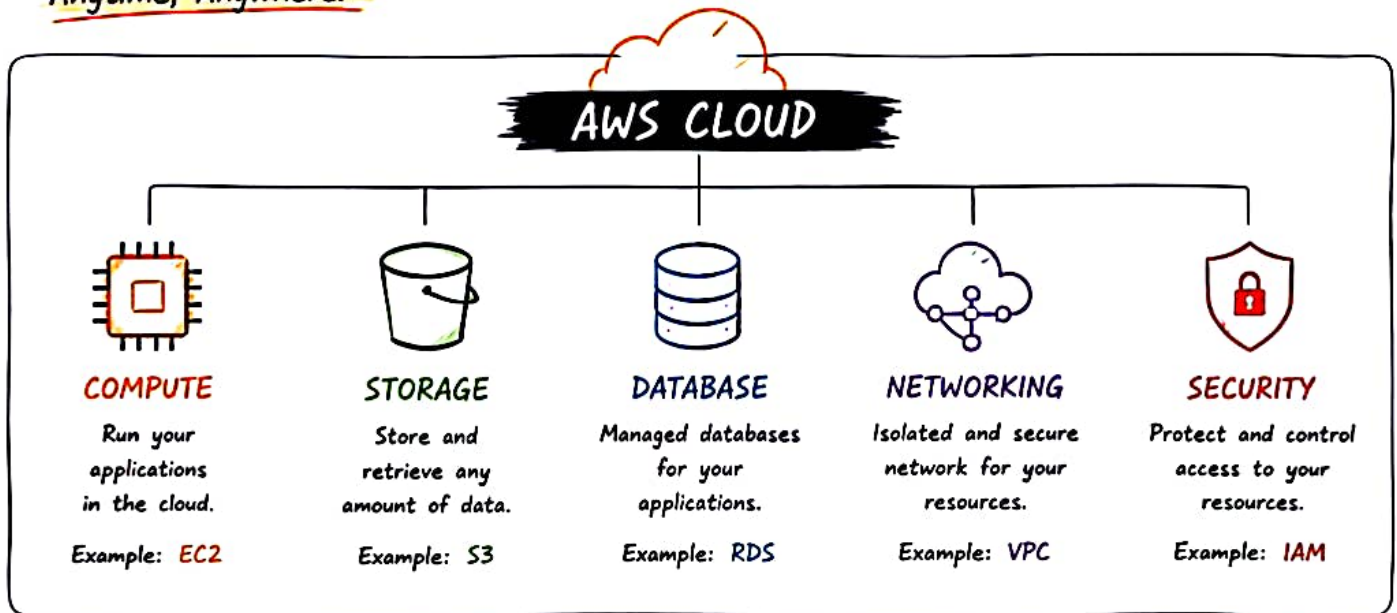
AWS FUNDAMENTALS

The Cloud That Powers the World

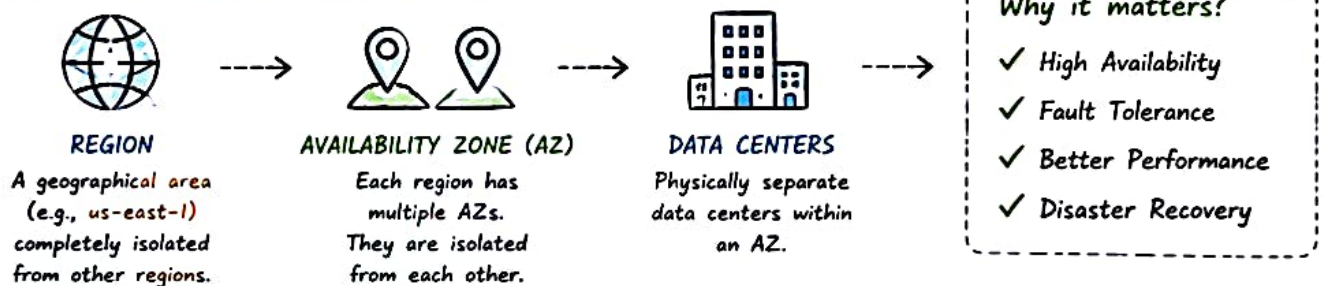
Build, deploy and scale any application -
Anytime, Anywhere.

66

AWS is a cloud platform that provides **on-demand** IT resources over the internet with **reliability, security and performance.**



AWS GLOBAL INFRASTRUCTURE



HOW IT WORKS (EXAMPLE)



Don't try to learn all AWS services at once.
First understand the **basics**. Then everything makes sense!



AWS

FUNDAMENTALS

02 | IDENTITY AND ACCESS MANAGEMENT (IAM)



WHY IAM?

IAM helps you securely control **who** can access AWS resources and **what** they can do.

The right people. The right access. The right resources.

THE BUILDING BLOCKS OF IAM



USERS

An entity with long-term credentials that represents a person or application.



GROUPS

A collection of users. Permissions are assigned to groups, not individual users.



ROLES

An IAM identity that you can assume to get temporary permissions.



POLICIES

JSON documents that define permissions (what actions are allowed or denied).

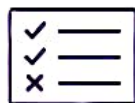
HOW IAM WORKS

1 AUTHENTICATE



The user or application signs in using credentials (password, access keys, MFA, etc.).

2 AUTHORIZE



IAM checks the policies attached to the user, group, or role.

3 PERMIT / DENY



IAM determines whether the action is allowed or denied.

4 ACCESS GRANTED



If allowed, the user can access and manage the AWS resources.

POLICY EFFECTS



ALLOW

Explicitly allows an action on a resource.



DENY

Explicitly denies an action. Deny overrides all allows.



IMPLICIT DENY

If no policy allows an action, it is implicitly denied.

EXAMPLE



Alice



Developers



EC2-Role

Policy

```
{
  "Effect": "Allow",
  "Action": ["s3:ListBucket",
    "s3:GetObject"],
  "Resource": "arn:aws:s3:::my-bucket/*"
}
```

Can list bucket and read objects from my-bucket in S3.



BEST PRACTICE

Use least privilege – give only the minimum permissions needed to perform the task. Nothing more.



REMEMBER: Strong security starts with strong identity and access management.

03 AWS

EC2

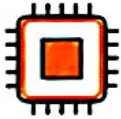
Elastic Compute Cloud

Virtual servers
in the cloud.
Launch.
Scale.
Done.



Why EC2?

- ✓ Full control
- ✓ Flexible & Scalable
- ✓ Pay only for what you use



Amazon EC2 provides resizable compute capacity in the cloud. You can launch virtual servers (*instances*) and manage them to run your applications.

KEY COMPONENTS OF EC2



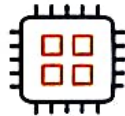
INSTANCE

A virtual server in the cloud. The basic building block.



AMI (AMAZON MACHINE IMAGE)

A template that contains the OS, applications & settings to launch an instance.



INSTANCE TYPE

Defines the CPU, memory, storage and network capacity of the instance.



EBS (ELASTIC BLOCK STORE)

Persistent storage for your instance. Data remains even if instance stops.



SECURITY GROUP

Acts as a virtual firewall that controls inbound and outbound traffic.

HOW AN EC2 INSTANCE CONNECTS



Internet



Security Group (Firewall)



EC2 Instance (Running)



EBS Volume (Storage)

IMPORTANT

By default, all inbound traffic is blocked. Only rules you add in Security Group are allowed.

INSTANCE TYPES (EXAMPLES)



t3 / t4g (General Purpose)
Balanced CPU, memory & networking. Good for most workloads.



m6i / m7i (Memory Optimized)
More memory for in-memory databases, big data, analytics.



c6i / c7i (Compute Optimized)
High CPU performance for compute intensive applications.



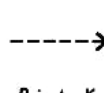
You can change instance type anytime by stopping the instance.

KEY PAIR (SSH ACCESS)

A key pair allows you to securely connect to your EC2 instance using SSH (Linux) or RDP (Windows).



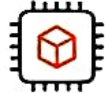
Your Machine



Private Key (.pem file)



Public Key (Stored on instance)



EC2 Instance

BEST PRACTICES

- ✓ Keep your .pem file safe. Do not share it.
- ✓ Use IAM roles instead of storing credentials on instances.
- ✓ Regularly update AMI and system packages.



EC2 gives you the power to run anything in the cloud.
Right instance. Right size. Right cost.

04 AWS S3

Simple Storage Service

Store any amount
of data.
Access anywhere.
Built for durability.



Why S3?

- ✓ Highly Durable (11 9's durability)
- ✓ Scalable & Available
- ✓ Secure
- ✓ Cost-effective
- ✓ Pay for what you use



Amazon S3 is object storage built to store and retrieve any amount of data from anywhere on the web. It is designed for 99.999999999% (11 9's) durability and 99.99% availability.

KEY CONCEPTS



BUCKET

A container that holds objects (files) in S3. Bucket names must be unique globally.



OBJECT

The data you store in S3. Consists of data and metadata (key, size, type, etc.).



KEY

The unique name (path) used to identify an object inside a bucket.



METADATA

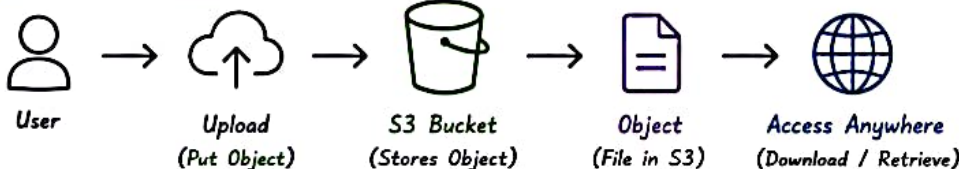
Data about your object. (System metadata) or custom metadata (user-defined).



ACCESS CONTROL

Manage who can access your bucket and objects using policies and permissions.

HOW S3 WORKS



BUILT TO LAST

- ✓ High Durability
- ✓ High Availability
- ✓ Automatic Replication
- ✓ Data Protection Options

S3 STORAGE CLASSES (COMMON)



S3 STANDARD

Frequently accessed data with low latency.



S3 STANDARD-IA

Infrequently accessed data. Lower storage cost.



S3 GLACIER

Archive data. Retrieval takes minutes to hours.



S3 DEEP ARCHIVE

Lowest cost for long term archive. Retrieval takes hours.

EXAMPLE USE CASES



Website
Hosting



Backup &
Recovery



Big Data
Analytics



Media &
Content



Logs &
Archiving

S3 SECURITY BASICS



Bucket Policy
Control access
at bucket level.



IAM Permissions
Manage access
using IAM users
and roles.



Encryption
Protect data at rest
(SSE-S3, SSE-KMS)
and in transit.



BEST PRACTICES

- ✓ Use meaningful bucket names.
- ✓ Enable Versioning to protect against accidental deletes.
- ✓ Enable Lifecycle rules to manage old data and reduce cost.
- ✓ Use IAM roles, not access keys.
- ✓ Enable encryption for sensitive data.
- ✓ Monitor access using CloudTrail and S3 logs.



“

S3 is more than storage.

It's the foundation of durable, scalable and secure data in the cloud.

”

05 AWS

RDS

Relational Database Service

Managed relational databases in the cloud.

Focus on your app, not database management.



Why RDS?

- ✓ Managed & Automated
- ✓ Highly Available
- ✓ Scalable Storage
- ✓ Secure
- ✓ Backups & Recovery
- ✓ Pay for what you use



Amazon RDS makes it easy to set up, operate, and scale a relational database in the cloud. It handles time-consuming database tasks like provisioning, patching, backups, monitoring, and scaling.

• KEY COMPONENTS •



DB INSTANCE

The database environment where your data is stored and managed.



ENGINE

Database engine you choose (MySQL, PostgreSQL, MariaDB, Oracle, SQL Server).



STORAGE

Persistent storage (EBS) that automatically scales with your needs.



BACKUPS

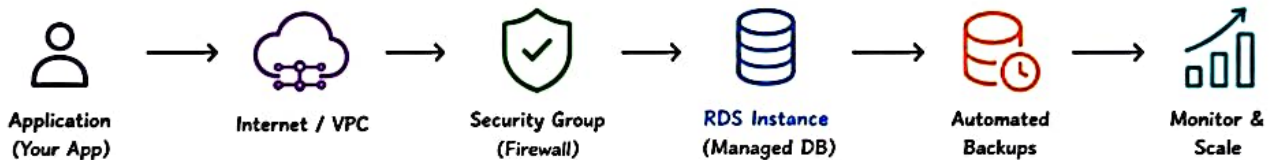
Automated backups and point-in-time recovery to protect your data.



SECURITY

VPC isolation, security groups, encryption, and IAM for secure access.

HOW RDS WORKS



RDS handles the heavy lifting so you can focus on building great applications.

SUPPORTED ENGINES



MySQL

Open-source, popular, and widely used.



PostgreSQL

Advanced open-source relational database.



MariaDB

MySQL-compatible, open-source.



Oracle

Enterprise-grade database.



SQL Server

Microsoft's enterprise database.

KEY FEATURES



High Availability

Multi-AZ deployments for automatic failover.



Scalability

Easily scale storage and compute as your workload grows.



Automated Backups

Regular snapshots and point-in-time recovery.



Monitoring

Built-in performance insights and metrics.



Automated Maintenance

Patching and updates with minimal downtime.

BEST PRACTICES



- ✓ Use **Multi-AZ** for production databases.
- ✓ Enable **automated backups** and test restore regularly.
- ✓ Use **IAM authentication** instead of hardcoded passwords.
- ✓ **Encrypt your data** at rest and in transit.
- ✓ **Monitor performance** and set alarms.
- ✓ **Right-size your instance** to optimize cost.



RDS gives you a **reliable, secure, and scalable** database in the cloud, so you can focus on building – **not managing**.

06 AWS VPC

Virtual Private Cloud

Your own private
network in the cloud.
Isolated. Secure.
Fully in your control.



Why VPC?

- ✓ Complete control over your network
- ✓ Logical isolation
- ✓ Secure & configurable
- ✓ Connect to on-premises networks
- ✓ Highly available applications



Amazon VPC lets you launch AWS resources in a logically isolated virtual network that you define. You have full control over your virtual networking environment, including IP address range, subnets, route tables and network gateways.

KEY COMPONENTS



VPC

A logically isolated section of the AWS Cloud where you can launch resources.



SUBNET

A range of IP addresses in your VPC. Subnets reside in a single Availability Zone.



ROUTE TABLE

Contains rules that determine where network traffic is sent.



INTERNET GATEWAY

Allows communication between resources in your VPC and the internet.



NAT GATEWAY

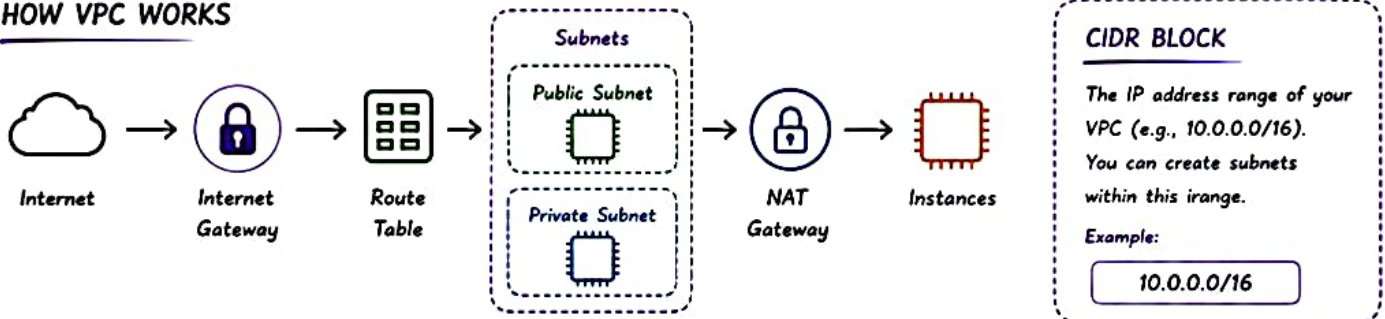
Enables instances in private subnets to access the internet outbound.



SECURITY GROUP

Acts as a virtual firewall to control inbound and outbound traffic to instances.

HOW VPC WORKS



SUBNET TYPES



PUBLIC SUBNET

Has a route to the Internet Gateway. Resources can be accessed from the internet.



PRIVATE SUBNET

No direct route to the internet. Use NAT Gateway for outbound internet access.



Use public subnets for resources that need internet access. Use private subnets for resources that should be private.

VPC PEERING & CONNECTIVITY



VPC Peering

Connect two VPCs privately using a peering connection.



VPN Connection

Secure connection between your VPC and on-premises network.



AWS Direct Connect

Dedicated network connection from your data center to AWS for consistent performance.

BEST PRACTICES

- ✓ Use private subnets for your databases and backends.
- ✓ Use Security Groups to allow only necessary traffic.
- ✓ Enable VPC Flow Logs for monitoring and troubleshooting.
- ✓ Use NAT Gateway instead of public IPs for outbound access.
- ✓ Plan your IP address range carefully.



VPC is the foundation of your **secure** and **scalable** architecture on AWS.
Design your network. Secure your applications. Scale with confidence.

07 AWS IAM

Identity and Access Management

The right people
in your organization
get the right access
to the right resources.
Nothing more.
Nothing less.



Why IAM?

- ✓ Centralized access management
- ✓ Secure your AWS resources
- ✓ Fine-grained permissions
- ✓ Multi-user access
- ✓ Compliance and audit support
- ✓ Free service



AWS IAM helps you securely control access to AWS services and resources. You can create users and groups in your organization and use permissions (policies) to allow or deny their access.

KEY CONCEPTS



USER

An individual identity with unique credentials that can access AWS.



GROUP

A collection of users. Permissions can be assigned to groups.



POLICY

A JSON document that defines permissions. Attached to users, groups, or roles.



ROLE

An identity that AWS services or trusted users can assume to access resources.



PERMISSION

Allows or denies access to perform actions on AWS resources.

HOW IAM WORKS



User
(People)



Groups
(Organize)



Policies
(Permissions)



Allow / Deny
Access



AWS Resources
(Services)

EXAMPLE

A developer needs access to S3 but not to EC2. Attach a policy that allows S3 actions and denies EC2 actions.

TYPES OF POLICIES



Managed Policies
AWS managed or customer managed. Reusable and easy to attach.



Inline Policies
Embedded directly into a user, group, or role.



Permissions Boundaries
Set the maximum permissions an identity can have.

BEST PRACTICES

- ✓ Follow the principle of least privilege.
- ✓ Use groups to manage permissions, not individual users.
- ✓ Use IAM roles for applications and AWS services.
- ✓ Enable MFA for all users.
- ✓ Regularly review and audit permissions.
- ✓ Use descriptive names for users, groups, and policies.

IMPORTANT



Root User
Has full access to your AWS account. Use it only for account setup and billing.



MFA
Multi-Factor Authentication adds an extra layer of security.



IAM is Global
IAM is a global service. It works across all regions in your account.



IAM is the gatekeeper of your AWS environment.
Secure your account. Control access. Protect your resources.

08 AWS LAMBDA

Serverless Compute Service

Run code **without** thinking about **servers**. Just upload your code and Lambda handles **everything**.



Why Lambda?

- ✓ No servers to manage
- ✓ Pay only for what you use
- ✓ Automatically scales
- ✓ Highly available
- ✓ Built-in integrations
- ✓ Focus on your code



AWS Lambda lets you run your code in response to events and automatically manages the compute resources for you. You pay only for the time your code executes, from **milliseconds** to **several minutes**.

KEY CONCEPTS



FUNCTION

A Lambda function is your code and its configuration.



TRIGGER

An event source that invokes your function (e.g., S3, API Gateway, DynamoDB, SNS).



RUNTIME

Lambda runs your code in a managed runtime (Python, Node.js, Java, etc.).



SCALING

Lambda automatically scales your function based on incoming requests.



PRICING

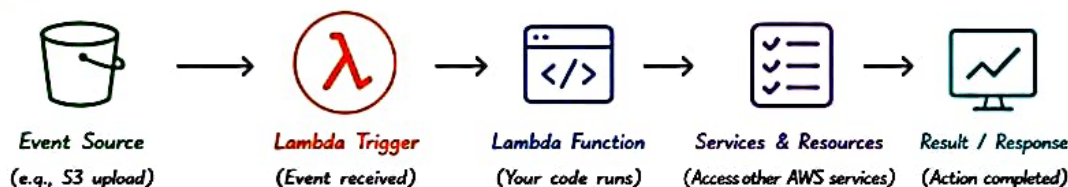
You pay only for the compute time you consume. No idle charges.



PERMISSIONS

IAM roles define what your function can access in AWS services.

HOW LAMBDA WORKS



EXAMPLE

When a file is uploaded to S3, Lambda is triggered, processes the file (e.g., resizes an image), and stores the output in another S3 bucket.

COMMON USE CASES

- File processing**
Process files uploaded to S3.
- Backend for APIs**
Run code in response to API requests.
- Data processing**
Process streams from DynamoDB, Kinesis, etc.
- Automations**
Send emails, notifications, or perform scheduled tasks.
- ML Inference**
Run machine learning models without managing servers.

KEY FEATURES

- Serverless**
No infrastructure to provision or manage.
- Automatic Scaling**
Handles thousands of requests simultaneously.
- Short-lived Execution**
Runs from milliseconds to 15 minutes per invocation.
- High Availability**
Built-in fault tolerance across multiple AZs.
- Built-in Integrations**
Works seamlessly with many AWS services.

BEST PRACTICES

- ✓ Keep functions small and focused.
- ✓ Use appropriate memory allocation to improve performance.
- ✓ Enable logging with CloudWatch Logs.
- ✓ Use environment variables for configuration.
- ✓ Handle errors and set timeouts appropriately.
- ✓ Use IAM roles with least privilege.



Lambda lets you focus on writing code. AWS handles the rest. **Code. Deploy. Scale. No servers.**



09 AWS SQS

Simple Queue Service

Decouple your applications.
Communicate asynchronously.
Scale with ease.



Why SQS?

- ✓ Decouple microservices & components
- ✓ Buffer and protect against spikes
- ✓ Highly scalable & reliable
- ✓ Message persistence
- ✓ Built-in retry & error handling
- ✓ Pay only for what you use



Amazon SQS is a fully managed message queuing service that enables you to decouple and scale microservices, distributed systems, and serverless applications. It acts as a reliable buffer between producers (senders) and consumers (receivers).

KEY CONCEPTS



QUEUE

A queue stores messages until they are processed by a consumer.



MESSAGE

The data unit that is sent to, stored in, and received from the queue.



PRODUCER

Sends messages to the queue. (Producers don't need to know who consumes.)



CONSUMER

Receives and processes messages from the queue. (Consumers don't know who sends.)



VISIBILITY TIMEOUT

The time a message remains invisible after being received by a consumer to prevent duplicate processing.



MESSAGE RETENTION

The duration a message is kept in the queue if not deleted (default 4 days, max 14 days).

HOW SQS WORKS



Producer

Sends message to SQS Queue



SQS Queue

Stores the message safely



Consumer

Receives and processes message



Message Deleted

After successful processing



If Failed

Message becomes visible again after visibility timeout

EXAMPLE

A user signs up → Message sent to SQS → Email service (consumer) picks it up and sends welcome email.



QUEUE TYPES



Standard Queue

- At-least-once delivery
- Best-effort ordering
- High throughput
- Default queue type



FIFO Queue

- Exactly-once processing
- First-In-First-Out order
- High throughput (with limits)
- Requires .fifo suffix

KEY FEATURES



Highly Durable

Messages are stored redundantly across multiple AZs.



Scalable

Automatically scales to handle millions of messages per second.



Retry & Redrive

Failed messages can be retried or moved to a Dead-Letter Queue.



Secure

IAM permissions, VPC endpoints, and server-side encryption.

COMMON USE CASES



Microservices Decoupling

Decouple services for better scalability and resilience.



Traffic Spikes

Buffer incoming requests during high traffic.



Background Tasks

Process long-running tasks (asynchronously).



Batch Processing

Collect and process data in batches.

BEST PRACTICES

- ✓ Use appropriate queue type (Standard or FIFO).
- ✓ Set visibility timeout based on processing time.
- ✓ Use Dead-Letter Queues for failed messages.
- ✓ Monitor with CloudWatch metrics and alarms.
- ✓ Use long polling to reduce empty responses.
- ✓ Keep messages small (up to 256 KB).



SQS helps your applications talk to each other without being tightly coupled.
Reliable. Scalable. Flexible.



10 AWS CLOUDWATCH

Monitor. Visualize. Act.

Full observability
of your AWS
resources and
applications.

Understand,
monitor, and act.



Why CloudWatch?

- ✓ Real-time monitoring and metrics
- ✓ Collect logs from any source
- ✓ Set alarms and take action
- ✓ Visualize performance
- ✓ Automate operational tasks
- ✓ Gain system-wide visibility
- ✓ Pay only for what you use



Amazon CloudWatch is a monitoring and observability service that provides data and actionable insights to monitor your applications, respond to system-wide performance changes, optimize resource utilization, and get a unified view of operational health.

KEY COMPONENTS



METRICS

Collects and tracks
metrics from AWS
resources and
custom sources.



LOGS

Collect, store, and
monitor log files
from AWS services,
applications, and
custom sources.



ALARMS

Set thresholds on
metrics and get
notified or trigger
actions when the
threshold is met.



DASHBOARDS

Create custom
dashboards to
visualize metrics
and logs in real
time.



EVENTS

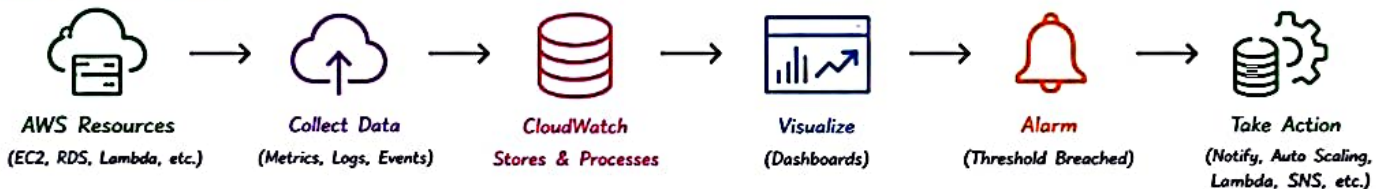
Detect state changes
in AWS resources
and route events
to targets.



INSIGHTS

CloudWatch Insights
helps you analyze
data and detect
anomalies.

HOW CLOUDWATCH WORKS



METRIC TYPES



Standard Metrics
AWS services publish
these metrics.



Custom Metrics
You publish your own
application metrics.



High-Res Metrics
At 1-second resolution
for detailed monitoring.



CloudWatch stores metrics
for 15 months.

ALARM ACTIONS



Amazon SNS
Send notifications



AWS Lambda
Run custom code



Auto Scaling
Scale your resources



Systems Manager
Run automation

LOG SOURCES



AWS Services
VPC Flow Logs, CloudTrail,
ELB Logs, etc.



Applications
Logs from your applications



On-Premises
Use CloudWatch Agent
to collect logs



Custom Sources
Push logs using APIs
or SDKs

BEST PRACTICES

- ✓ Use meaningful metrics and dimensions.
- ✓ Set appropriate alarm thresholds.
- ✓ Use dashboards for quick visual insights.
- ✓ Enable detailed monitoring for critical resources.
- ✓ Use log retention policies to manage costs.
- ✓ Review and act on alarms promptly.



CloudWatch gives you the visibility you need to run your applications
reliably and efficiently in the cloud.

Monitor everything. Understand anything. Improve continuously.

